

ARJUN VARMA

(347) 987-9427 | av3342@columbia.edu | linkedin.com/in/varma-arjun | github.com/arjunvarma2000 | arjun-varma.com

EDUCATION

Columbia University

Master of Science in Data Science

Teaching Assistant, Columbia Business School: Business Analytics II (Foundations of AI) and Hollywood and Big Data

New York, NY

Expected Dec 2026

Vellore Institute of Technology

Bachelor of Technology in Electronics and Communication Engineering

Special Achiever Award | Merit Scholarship

Vellore, IN

May 2022

WORK EXPERIENCE

Novo Nordisk

Plainsboro, NJ

Data Scientist Intern

Jun 2026 - Present

- Equipped field teams with explainable recommendations by improving propensity scoring and next-best-engagement models and building an LLM layer for decision support across multi-source behavioral, access, competitive, and adoption signals.
- Prioritized ~500 of 10,000+ providers nationwide by integrating unmet-need, market-opportunity, and early-adoption signals from longitudinal claims.
- Achieved 5.2x top-decile lift with an XGBoost propensity model using provider profile, prescribing, access, and engagement features; combined predictions with rule-based signals to produce explainable priority tiers and a nationwide planning dashboard.

ZS Associates

Pune, IN

Advanced Data Science Associate Consultant

Feb 2022 - Jun 2025

- Recovered missing categorical labels in sparse medical transaction data with positive-unlabeled classification, expanding analytical coverage from ~40% to 95% across segments and territories.
- Productionized the models with MLflow tracking, feature/prediction drift monitoring, and CI validation, reducing silent failures.
- Reduced weekly decision cycles from days to minutes for 100+ stakeholders across a \$10B portfolio by translating 5+ high-volume sources into a reusable Spark/SQL data warehouse, product KPIs, and self-service analyses.
- Raised first-pass analytical quality above 99% and saved ~50 hours per month by leading a five-member modernization of legacy business rules into validated, reusable logic.
- Established audit-ready real-world evidence (RWE) modeling and segmentation across \$4B+ in products by designing cohort and entity features over millions of incomplete and miscoded records.

PROJECT EXPERIENCE

DEUCE: Live Tennis Forecasting Platform - Python, XGBoost, Next.js, GitHub Actions

[GitHub](#) / [Live Demo](#)

- Powered live match forecasting and tournament simulation from 87,957 evaluated ATP/WTAs matches by combining surface Elo, opponent-adjusted serve/return Markov features, and XGBoost.
- Improved leakage-free walk-forward Brier score from 0.2006 to 0.1950 (ATP) and from 0.2114 to 0.2017 (WTA); published frozen pre-match forecasts, model scorecards, and market comparisons.
- Operate live forecasts through hourly multi-source data refreshes, daily retraining, schema/freshness gates, failure alerts, Python/TypeScript CI, and automated Firebase deployment.

GAFFER: Live World Cup Forecasting Platform - Python, pandas, NumPy, Next.js, GitHub Actions

[GitHub](#) / [Live Demo](#)

- Generated live World Cup forecasts from ~49K international matches by combining Elo, a time-weighted Dixon-Coles goal model, squad-value blending, and 50K-run Monte Carlo simulations.
- Achieved 0.887 three-outcome log loss versus a 1.05 baseline across 8,136 walk-forward matches since 2018 and confirmed the squad-value blend improved out-of-sample performance on a 2024-2026 holdout.
- Keep forecasts current, benchmarked, and auditable through hourly refreshes, immutable pre-match predictions, Brier/log-loss and Kalshi comparisons, material-change gates, automated tests, and Firebase delivery.

Multi-Agent Analytics Copilot - Python, LangChain, FastAPI, DuckDB, Next.js

[GitHub](#) / [Live Demo](#)

- Converted natural-language questions into validated SQL, visualizations, and narrated answers with row-level citations through planner, SQL, validator, chart, and narrator agents.
- Improved query reliability and portability across DuckDB, Postgres, and Snowflake through SQL dry runs, row-count/type audits, self-critique with retries, regression evaluations, and per-query latency/cost tracking.

Filing Intelligence: Financial Research RAG - Python, FastAPI, ChromaDB, Next.js, GCP

[GitHub](#) / [Live Demo](#)

- Produced cited financial research across 15 companies and 4,967 passages from 128 filings, earnings decks, and call transcripts by building an evidence-first RAG workspace with scoped Chroma retrieval, deterministic ranking, and bounded context.
- Sustain reliable, auditable retrieval with page/line citations and 72 automated tests spanning retrieval, authentication, API contracts, and production builds; operate authenticated FastAPI/Next.js services, source-page viewers, and freshness checks.

SKILLS

Programming & ML: Python, SQL, R; pandas, NumPy, scikit-learn, PyTorch, XGBoost, SHAP

Modeling & Experimentation: Propensity scoring, A/B testing, experiment design, hypothesis testing, causal inference; ranking/recommendation, probabilistic forecasting, Monte Carlo simulation

Evaluation & Monitoring: Walk-forward validation, Brier score, multiclass log loss, regression suites, feature/prediction drift, data quality validation

Applied AI: LLMs/RAG, LangChain, agentic workflows, multi-agent orchestration, LLM evaluation and guardrails, FastAPI, REST APIs

Data & MLOps: Spark/PySpark, Databricks, MLflow, production ML, ETL/ELT, data warehousing, Docker, CI/CD, GitHub Actions

Databases & Cloud: Postgres, Snowflake, DuckDB, ChromaDB; AWS (S3, EMR, Athena, SageMaker); GCP Cloud Run